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Example:-

Find bases for the eigen spaces

and find a matrix P that diagonalizes.Solution:-The characteristic equation of A is

$$\det(\lambda I - A) = \det\left(\begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}\right) = 0$$

$$\Rightarrow \begin{vmatrix} \lambda & 0 & 2 \\ -1 & \lambda-2 & -1 \\ -1 & 0 & \lambda-3 \end{vmatrix} = 0$$

$$\Rightarrow \lambda \begin{vmatrix} \lambda-2 & -1 \\ 0 & \lambda-3 \end{vmatrix} - 0 + 2 \begin{vmatrix} -1 & \lambda-2 \\ -1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow \lambda(\lambda-2)(\lambda-3) + 2(0 - (-1)(\lambda-2)) = 0$$

$$\Rightarrow (\lambda^2 - 2\lambda)(\lambda-3) + 2(\lambda-2) = 0$$

$$\Rightarrow \lambda^3 - 2\lambda^2 - 3\lambda^2 + 6\lambda + 2\lambda - 4 = 0$$

$$\Rightarrow \lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

(2)

By using synthetic division.

$$\begin{array}{r|rrrr}
 1 & 1 & -5 & 8 & -4 \\
 & & 1 & -4 & 4 \\
 \hline
 & 1 & -4 & 4 & \boxed{0}
 \end{array}$$

$$\lambda^2 - 4\lambda + 4 = 0$$

$$\lambda^2 - 2\lambda - 2\lambda + 4 = 0$$

$$\lambda(\lambda - 2) - 2(\lambda - 2) = 0$$

$$(\lambda - 2)(\lambda - 2) = 0$$

$$\lambda - 2 = 0, \quad \cancel{\lambda - 2} \lambda - 2 = 0$$

$$\lambda = 2, \quad \lambda = 2$$

Thus distinct eigenvalues of A are

$$\lambda = 1, \quad \lambda = 2$$

So, there are two eigenspaces of A .

(3)

By definition,

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

is an eigenvector of A corresponding to λ if and only if x is non-trivial solution of

$$(\lambda I - A)x = 0$$

$$\begin{bmatrix} \lambda & 0 & 2 \\ -1 & \lambda-2 & -1 \\ -1 & 0 & \lambda-3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \textcircled{1}$$

case-I

$$\lambda = 2$$

$$\begin{bmatrix} 2 & 0 & 2 \\ -1 & 0 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$2x_1 + 0 \cdot x_2 + 2x_3 = 0$$

$$2x_1 = -2x_3$$

$$x_1 = -x_3$$

Let

$$\therefore x_3 = s$$

Then

$$x_1 = -s$$

again let

$$x_2 = t$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -s \\ t \\ s \end{bmatrix} = \begin{bmatrix} -s \\ 0 \\ s \end{bmatrix} + \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix}$$

$$= s \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Hence

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Case-II

from eq 4 ①

⑤

$$\lambda = 1$$

$$\begin{bmatrix} 1 & 0 & 2 \\ -1 & -1 & -1 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + 2x_3 = 0$$

$$-x_1 - x_2 - x_3 = 0$$

$$x_1 = -2x_3$$

$$x_1 + x_2 + x_3 = 0$$

Let

$$x_3 = S$$

$$x_2 = -x_1 - x_3$$

$$x_1 = -2S$$

$$x_2 = 2S - S$$

$$x_2 = S$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2S \\ S \\ S \end{bmatrix} = S \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

Hence

$$\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

There are three basis vectors in total.

So the matrix.

$$P = \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Now, find the inverse of P .

$$[P|I] = \left[\begin{array}{ccc|ccc} -1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_1 \rightarrow -R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 \end{array} \right]$$

$$R_3 \rightarrow -R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & -1 \end{array} \right]$$

$$R_1 - 2 \cdot R_3$$

and

$$R_2 - R_3$$

Final matrix:-

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & -1 & 0 & -1 \end{array} \right]$$

Hence

$$P^{-1} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix}$$

Diagonalizes A, As a check
verify that

$$P^{-1}AP = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$P^{-1}AP = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$